The Chemical Engineering faculty at IIT Bombay are collectively working on an exceptionally broad and deeply specific array of research topics, including:

- Design and synthesis of lower-scaling coupled-cluster quantum-chemical methods (EOM-CCSD, similarity-transformed second-order approximate CC) for accurate prediction of electron affinities, ionization potentials, and X-ray spectra of biomolecular subunits, and development of pair-natural-orbital equation-of-motion CC methods for core-binding forces;

- Many-body quantum-chemistry innovations such as dressed coupled-cluster theories with implicit triple excitations for hydrogen-bonded systems, nonlinear-dynamics–driven coupled-cluster acceleration via supervised machine learning, and synergetics-inspired approximate CC theories with adiabatic decoupling;

- Ultrafast molecular dynamics and fluorescence spectroscopy, including pump-probe upconversion studies of plasmon-enhanced dye aggregate disruption, single-particle carrier-replenishment kinetics in quantum dots, selective heavy-metal sensing mechanisms via photoluminescent quantum dots, and second-harmonic generation for semiconductor/dielectric interface characterization;

- Continuous-flow polymerization and conducting polymers, featuring high-throughput, template-free continuous flow synthesis of polyaniline nanofibers, supramolecular alignment yielding defect-free polymers with ultra-high mobility, ITO-free flexible electrochromic devices, and in situ quantum-dot encapsulation to boost photoconductivity in polymer composites;

- Glycomics and one-pot carbohydrate synthesis, including total syntheses of conjugation-ready Pseudomonas aeruginosa O11 and Staphylococcus aureus Type 5 capsular trisaccharides, protecting-group innovations for oligosaccharide assembly, core trisaccharide syntheses of lipoteichoic acids, and dolichol-phosphate intermediate construction for N-glycosylation studies;

- Molecular-level interfacial electrochemistry for energy conversion, such as acid-rinsed Li-rich oxide cathode stability, redox-center mapping in lithium intercalation, free-energy regime analysis for CO_2 reduction screening, and design of Co-free high-voltage cathodes, Na–S batteries, and fluoride-rich interlayers in anodeless solid-state cells;

- Process intensification and real-time control in metal AM and joining, including closed-loop GMA-brazing waveform control, coupled heat–fluid simulations in laser powder-bed fusion, in-situ heat-input monitoring, and powder-bed AM across diverse alloy systems;

- Photo(electro)chemical solar fuels, featuring defect-engineered GaN/ZnO heterojunction photoanodes with surface-platinized cocatalysts, ceria-based redox ceramics for solar-thermochemical methane reforming, high-pressure PEM electrolyzers with Ti–Nb alloy plates for >2 A/cm² at <1.8 V, and dynamic control of polymer electrolyte fuel cells for grid stabilization;

- Smart-grid and microgrid optimization, using multi-objective evolutionary algorithms for PV–wind–battery–supercapacitor–pumped hydro integration, droop-less virtual synchronous generator control, adaptive fuzzy wavelet harmonic compensation, and physics-informed neural-network digital twins trained on PMU data;
- Carbon capture and utilization, with amine-functionalized MOFs having hierarchical pores for >4 mol CO_2/kg at 0.15 bar, bipolar electrochemical direct air-capture cells with humidity-swing sorbents, and biochar from agricultural residues via microwave pyrolysis with in-situ syngas reforming;
- Advanced combustion science, including OH-PLIF mapping of ultra-lean swirling methane–air flames under high pressure, acoustically forced atomization studies for ammonia–cofired turbines, and skeletal chemical-kinetics simulations with adaptive mesh refinement for supersonic combustion;
- Techno-economic modeling of renewable hydrogen export, integrating GIS-based resource mapping, electrolyzer-siting optimization, and logistics costing for liquid organic hydrogen carriers;
- Life-cycle assessment using region-specific inventories and consequential LCA under dynamic demand scenarios;
- Microstructured reactor design for Fischer–Tropsch synthesis with gradient-doped Co/Fe catalysts, coupled membrane-permeation water removal;
- AI-enabled demand-response forecasting via graph-based spatio-temporal deep learning on smart-meter data;
- High-temperature thermal storage using molten salts stabilized with nano-oxide additives and PCM-enhanced packed beds;
- Hydrogen blending in gas networks, studying pipeline embrittlement via acoustic-emission-monitored stress analyses of welded joints;
- Rotary heat exchangers employing phase-change porous matrices for industrial waste-heat recovery, optimized through adjoint topology optimization.

This superset encapsulates the precise, current and recent research endeavors across the Chemical Engineering faculty at IIT Bombay, ranging from quantum-chemical method development through large-scale energy systems and instrumentation, providing a comprehensive map of student-involved projects and cutting-edge investigations.